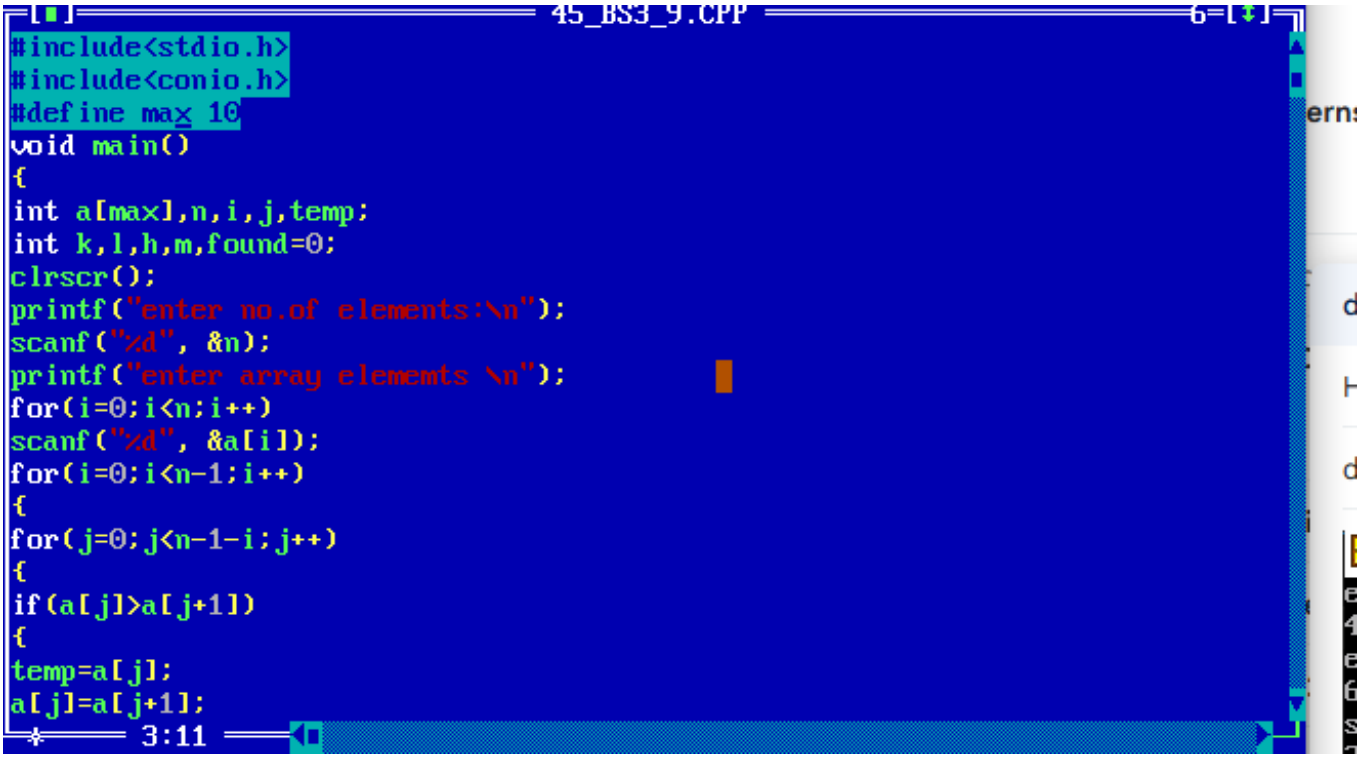
Implementation of Binary Search Technique

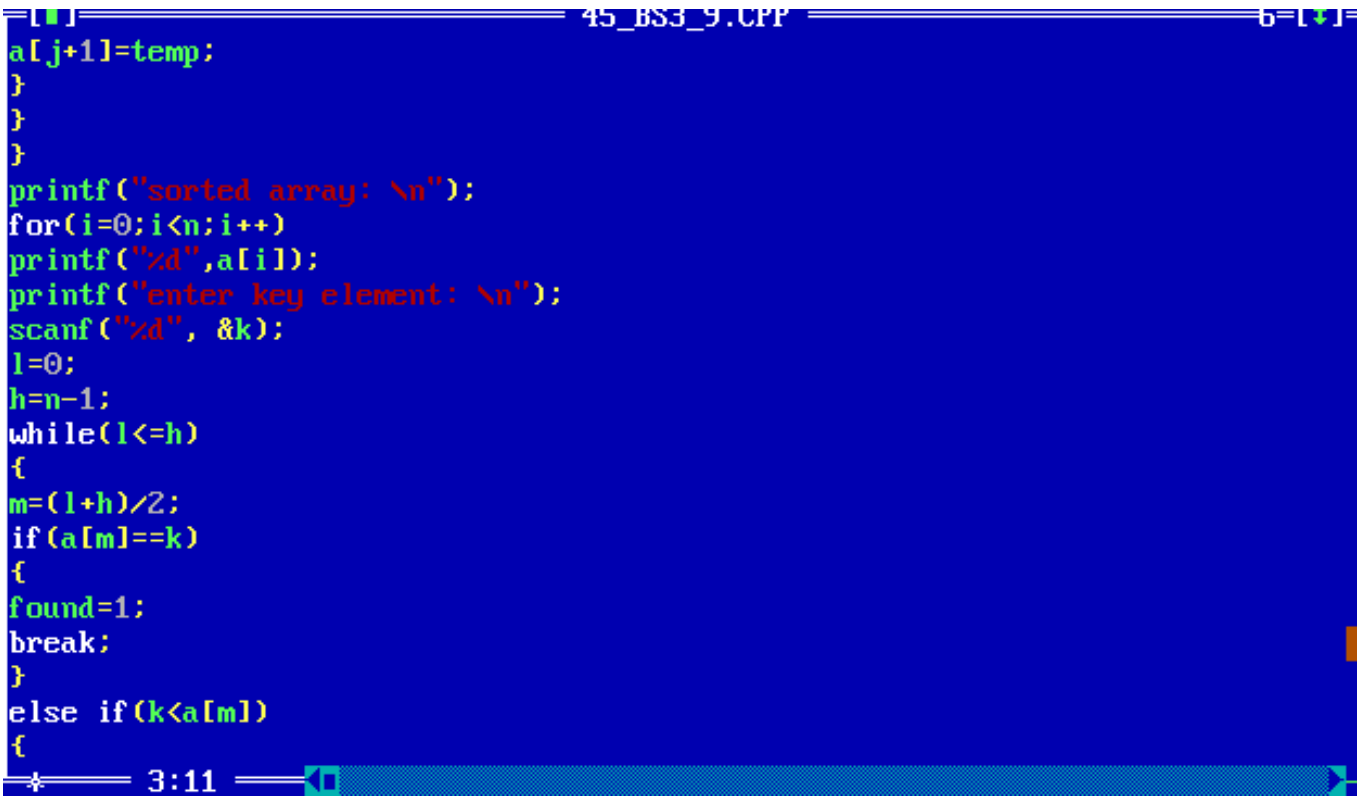
Aim

To implement the Binary Search technique to search for an element in a sorted array.

Program

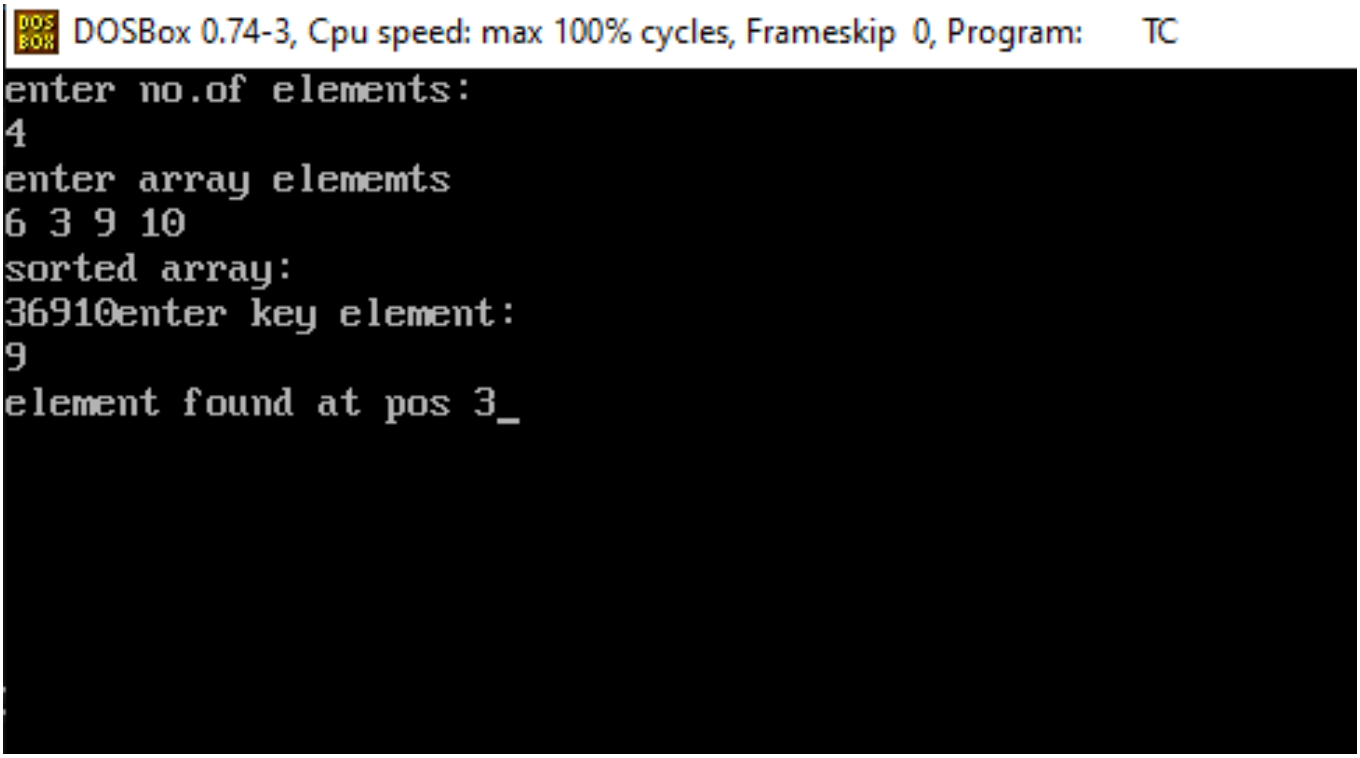


```
#include<stdio.h>
#include<conio.h>
#define max 10
void main()
{
int a[max],n,i,j,temp;
int k,l,h,m,found=0;
clrscr();
printf("enter no.of elements:\n");
scanf("%d", &n);
printf("enter array elememts \n");
for(i=0;i<n;i++)
scanf("%d", &a[i]);
for(i=0;i<n-1;i++)
{
for(j=0;j<n-1-i;j++)
{
if(a[j]>a[j+1])
{
temp=a[j];
a[j]=a[j+1];
```



```
a[j+1]=temp;
}
}
}
printf("sorted array: \n");
for(i=0;i<n;i++)
printf("%d",a[i]);
printf("enter key element: \n");
scanf("%d", &k);
l=0;
h=n-1;
while(l<=h)
{
m=(l+h)/2;
if(a[m]==k)
{
found=1;
break;
}
else if(k<a[m])
{
```

Output


DOSBox 0.74-3, Cpu speed: max 100% cycles, Frameskip 0, Program: TC
enter no.of elements:
4
enter array elememts
6 3 9 10
sorted array:
36910enter key element:
9
element found at pos 3_